

INTRODUCTION

Doppler's Effect is the change in the frequency (or pitch) of a wave as perceived by an observer when there is **relative motion** between the source and the observer.

It applies to sound, light, and other waves, and is widely used in science and technology.

Who Discovered It?



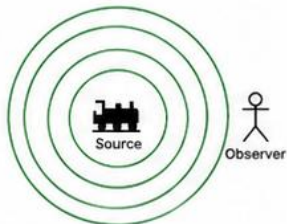
Christian Doppler
Austrian Physicist

1842

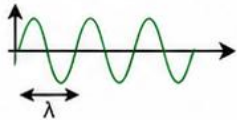
HOW DOES IT WORK?

1. Source & Observer at Rest

No change in frequency.

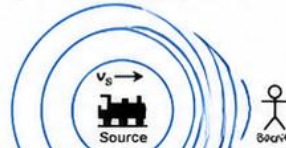


$$f = v/\lambda$$

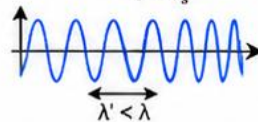


2. Source Moves Toward Observer

Waves get compressed →
Higher frequency (higher pitch).

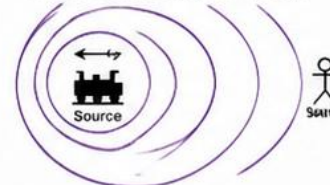


$$f' = \frac{v}{v - v_s} f$$

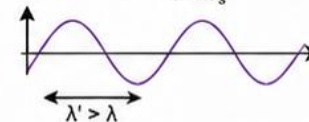


3. Source Moves Away from Observer

Waves get stretched →
Lower frequency (lower pitch).



$$f' = \frac{v}{v + v_s} f$$



KEY POINTS



Approaching
→ Higher Pitch



Receding
→ Lower Pitch



No Relative Motion
→ No Change



Applies to
Sound, Light
& Other Waves

REAL-WORLD EXAMPLES



Ambulance Siren

High pitch when approaching,
low when moving away.



Train Whistles

Pitch changes as train approaches
and passes.



Radar Guns (Police)

Measure the speed of
moving vehicles using
Doppler radar.



Weather Forecasting

Detects storms, wind
direction, and rainfall
using Doppler radar.

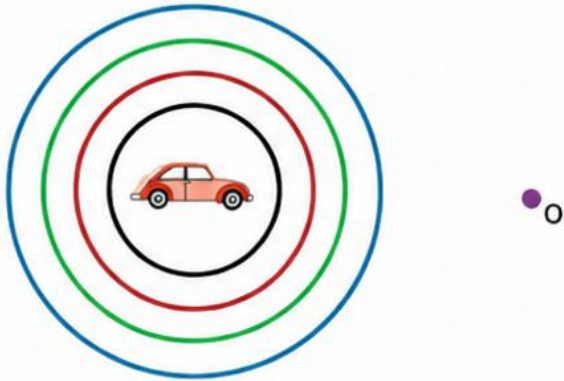


Medical Field

Measures blood flow and
detects heart issues
using Doppler ultrasound.

CASE I:

1. Both Source and Observer are at Rest.



Source and observer at rest

Here, both the source and the observer are at rest with respect to the medium.

The wavefronts reach the observer at the same rate as they are produced.

So, the observer hears the **true frequency**, with no change in pitch.

Apparent frequency heard by observer is,

$$f' = \frac{v'}{\lambda_e} \dots \dots \dots (i)$$

Here, $v' = v$, so equation (i) can be written as,

$$f' = \frac{v}{\lambda_e}$$

$$f' = \left(\frac{v}{\frac{v}{f}} \right) f \quad [\because \lambda_e = v/f]$$

$$f' = f$$

V = Velocity of sound

v_o = Velocity of observer

v_s = Velocity of source

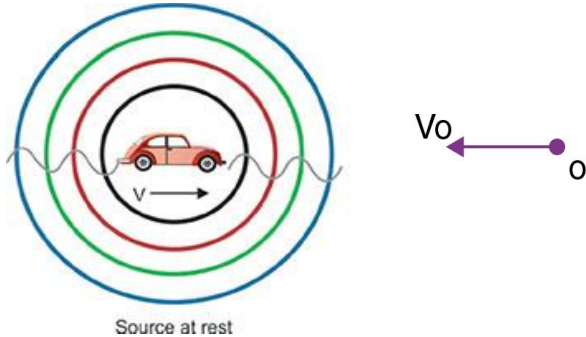
f = Original frequency of sound

f' = Apparent frequency

λ_e = Effective wavelength

CASE II:

1. Observer moves towards stationary source.



Here, the observer is moving toward the source. The observer meets the sound waves faster than normal.

This causes the observer to hear a **higher frequency**, even though the source frequency remains the same.

Apparent frequency heard by observer is,

$$f' = \frac{v'}{\lambda} \dots \dots \dots (i)$$

Here, $v' = v + v_0$, so equation (i) can be written as,

$$f' = \frac{v + v_0}{\lambda}$$

$$f' = \left(\frac{v + v_0}{v} \right) f \quad [\because v = \lambda f]$$

V = Velocity of sound

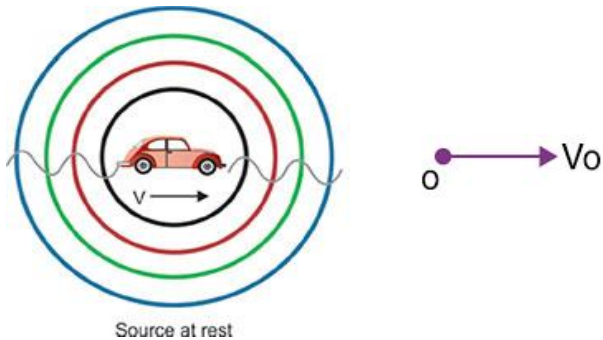
v₀ = Velocity of observer

f = Original frequency of sound

f' = Apparent frequency

CASE III:

1. Observer moves away from stationary source.



Here, the source is stationary, and the observer is moving away from the source.

Since the observer is moving away, they meet the sound waves less frequently than normal.

This causes the apparent **frequency to decrease**, even though the source frequency remains the same.

Apparent frequency heard by observer is,

$$f' = \frac{v'}{\lambda} \dots \dots \dots (ii)$$

Here, $v' = v - v_0$, so equation (ii) can be written as,

$$f' = \frac{v - v_0}{\lambda}$$

$$f' = \left(\frac{v - v_0}{v} \right) f \quad [\because v = \lambda f]$$

V = Velocity of sound

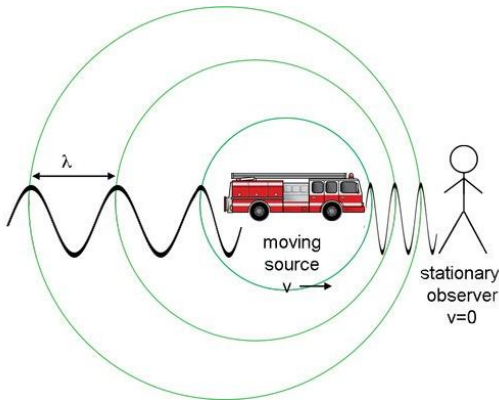
v₀ = Velocity of observer

f = Original frequency of sound

f' = Apparent frequency

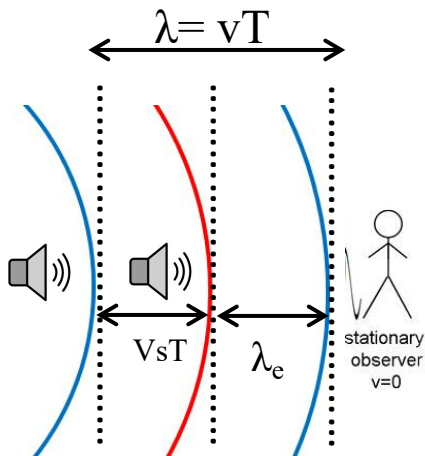
CASE IV:

1. Source moves towards stationary observer.



When the sound source moves towards a stationary observer, it creates compressed sound waves in front of it.

Because these waves are closer together, the observer hears a **higher frequency** than the actual frequency produced by the source.



Apparent frequency heard by stationary observer is,
 $f' = \frac{v}{\lambda_e} \dots \dots \dots (i)$

From fig:

$$\lambda_e = vT - v_s T$$

$$\lambda_e = \frac{v - v_s}{f}$$

Then,

$$f' = \left(\frac{v}{v - v_s} \right) f$$

λ_e = Effective wavelength

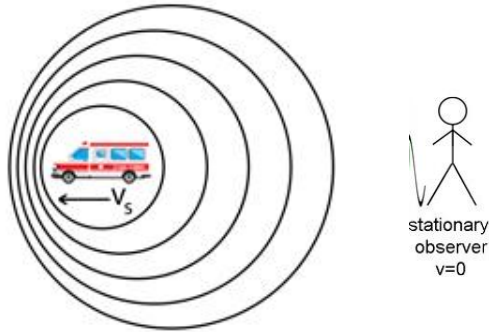
T = Time period

v_s = Velocity of source

v = Velocity of sound

CASE V:

1. Source moves away from stationary observer.



When the sound source moves away from a stationary observer, the sound waves spread out and the observer receives fewer waves per second. As a result, the observer hears a **lower apparent frequency** than the actual frequency.

Apparent frequency heard by stationary observer is,

$$f' = \frac{v}{\lambda_e} \dots \dots \dots (i)$$

Similarly,

$$\lambda_e = \frac{v + v_s}{f}$$

Then,

$$f' = \left(\frac{v}{v + v_s} \right) f$$

λ_e = Effective wavelength

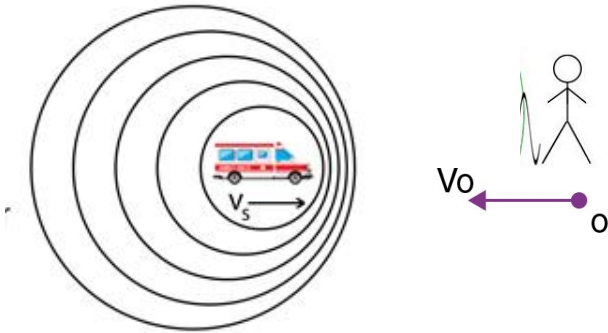
T = Time period

V_s = Velocity of source

V = Velocity of sound

CASE VI:

1. Both source and observer move towards each other.



When both the source and the observer move toward each other, the observer meets sound waves much faster than normal. The waves become more compressed, so the apparent frequency becomes **much higher** than the actual frequency.

When the source moves **toward** the observer it compresses the sound waves, reducing the wavelength,

$$\lambda_e = vT - v_s T$$

$$\lambda_e = \frac{v - v_s}{f}$$

The observer moves toward the incoming waves, so effective speed of waves relative to observer,

$$v' = v + v_o$$

Apparent frequency heard,

$$f' = \frac{v'}{\lambda_e}$$

Substituting values,

$$f' = \left(\frac{v + v_o}{v - v_s} \right) f$$

V = Velocity of sound

v_o = Velocity of observer

V_s = Velocity of source

F = Original frequency of sound

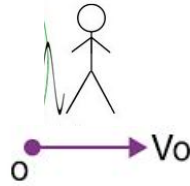
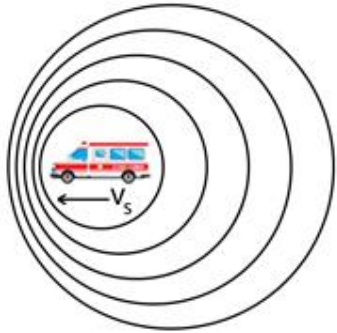
f' = Apparent frequency

λ_e = Effective wavelength

T = Time period

CASE VII:

1. Both source and observer move away from each other.



When both the source and the observer move away from each other, the observer meets sound waves slower than normal. The waves become more **stretched out**, so the apparent frequency becomes **lower than the actual frequency**.

When the source moves **away** from the observer, it **stretches** the sound waves, **increasing the wavelength**,

$$\lambda_e = vT + v_s T$$

$$\lambda_e = \frac{v + v_s}{f}$$

The observer moves **away from** the incoming waves, so the effective speed of waves relative to the observer **decreases**, $v' = v - v_o$

Apparent frequency heard,

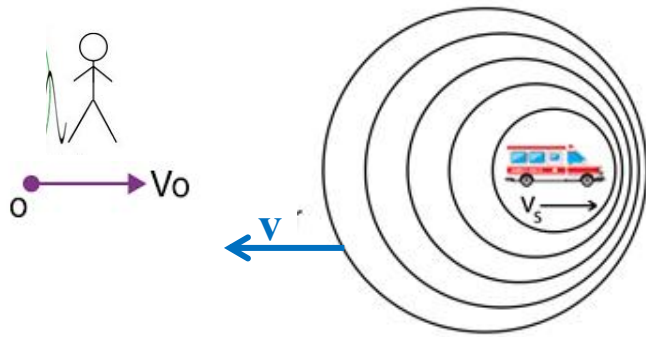
$$f' = \frac{v'}{\lambda_e}$$

Substituting values,

The apparent frequency of sound is, $f' = \left(\frac{v - v_o}{v + v_s} \right) f$

CASE VIII:

1. When the source is followed by the observer.



When the source moves away while the observer moves towards it, In such case sound wave and source move in the opposite direction and, the sound wave and observer also move in opposite direction.

The observer moves toward the incoming waves, so effective speed of waves relative to observer,

$$v' = v + v_0$$

and the apparent wavelength,

$$\lambda_e = \frac{v + v_s}{f}$$

Apparent frequency heard,

$$f' = \frac{v'}{\lambda_e}$$

Substituting values,

The apparent frequency of sound is,

$$f' = \left(\frac{v + v_0}{v + v_s} \right) f$$

V = Velocity of sound

v_0 = Velocity of observer

V_s = Velocity of source

F = Original frequency of sound

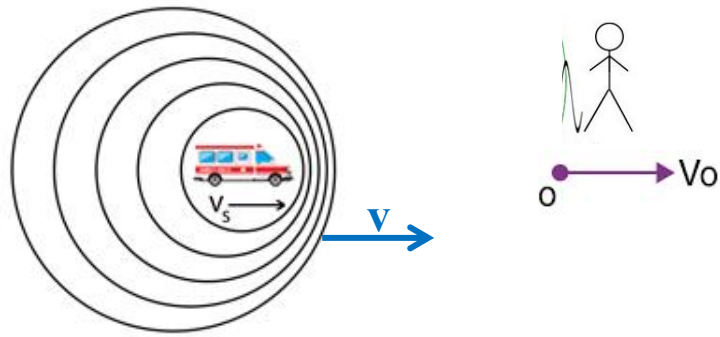
f' = Apparent frequency

λ_e = Effective wavelength

T = Time period

CASE IX:

2. When the observer is followed by the source.



When the observer moves away while the source moves towards it, In such case sound wave and source move in the same direction and, the sound wave and observer also move in same direction.

The observer moves **away from** the incoming waves, so the effective speed of waves relative to the observer **decreases**, $v' = v - v_o$

and the apparent wavelength,

$$\lambda_e = \frac{v - v_s}{f}$$

Apparent frequency heard,

$$f' = \frac{v'}{\lambda_e}$$

Substituting values,

The apparent frequency of sound is, $f' = \left(\frac{v - v_o}{v - v_s} \right) f$

v = Velocity of sound

v_o = Velocity of observer

v_s = Velocity of source

f = Original frequency of sound

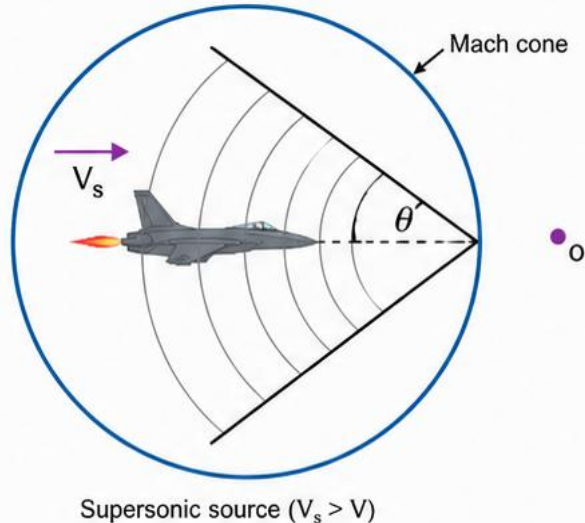
f' = Apparent frequency

λ_e = Effective wavelength

T = Time period

CASE X:

10. Sonic Boom (Source moves faster than speed of sound).



Here, the source moves faster than the speed of sound ($V_s > V$).

It outruns its wavefronts, which pile up along a cone called the Mach cone.

An observer experiences a sudden, loud “**boom**” as the cone passes.

Apparent frequency (as $V_s \rightarrow V$, $f' \rightarrow \infty$ in Case 4),

$$f' = \frac{v'}{\lambda_e} \dots\dots\dots (i)$$

From Case 4 : $f' = \frac{V}{V - V_s} f$

For $V_s > V$, the wavefront model breaks down. Instead, the half-angle of Mach cone is,

$$\sin \theta = \frac{V}{V_s} \quad [\because V_s > V]$$

$$\theta = \arcsin \left(\frac{V}{V_s} \right)$$

$$\text{Mach number, } M = \frac{V_s}{V} \quad (M > 1)$$

V = Velocity of sound

V_s = Velocity of source ($V_s > V$)

v_o = Velocity of observer (here $v_o = 0$)

f = Original frequency of sound

f' = Apparent frequency

λ_e = Effective wavelength

θ = Half-angle of Mach cone

M = Mach number ($M > 1$)

REAL-LIFE EXAMPLES & APPLICATIONS

1. Ambulance Siren:

When an ambulance comes toward us, the siren sounds high-pitched, and when it moves away, the pitch decreases.



2. Train Whistles:

The sound of a moving train changes as it approaches and passes by a station.



3. Radar Guns (Police):

Doppler radar is used by traffic police to measure the speed of moving vehicles.



4. Weather Forecasting:

Doppler radar is used to detect storms, wind direction, and rainfall.



5. Medical Field:

Doppler ultrasound is used to measure the flow of blood in arteries and detect heart issues.



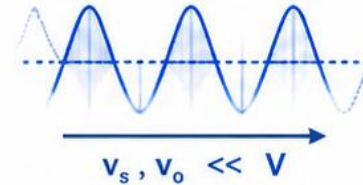
LIMITATIONS OF DOPPLER'S EFFECT



01

Assumes Low Speed:

The formulas work accurately only when the speeds of the source and observer are much smaller than the speed of sound.



02

Medium Required:

Sound Doppler Effect cannot occur in a vacuum because sound needs a medium.



03

Wind Effect:

The presence of wind can change the speed of sound and cause errors.



04

Complex Motion:

If the source or observer moves in a curved path instead of a straight line, the calculations become complicated.





*Thank
You!*

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FOR YOUR ATTENTION

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PREPARED BY

KISMAT DAHAL
— — — — —